

Table 1: Common Laplace transform pairs

Item no.	$f(t)$	$F(s)$
1.	$\delta(t)$	1
2.	$u(t)$	$\frac{1}{s}$
3.	$tu(t)$	$\frac{1}{s^2}$
4.	$t^n u(t)$	$\frac{n!}{s^{n+1}}$
5.	$e^{-at}u(t)$	$\frac{1}{s+a}$
6.	$\sin \omega t u(t)$	$\frac{\omega}{s^2 + \omega^2}$
7.	$\cos \omega t u(t)$	$\frac{s}{s^2 + \omega^2}$

Q1. Given the following differential equation, solve for $y(t)$ if all initial conditions are zero. Use the Laplace transform.

$$\frac{d^2 y}{dt^2} + 12 \frac{dy}{dt} + 32y = 32u(t)$$

Q2. Find the transfer function,

$$G(s) = \frac{C(s)}{R(s)},$$

corresponding to the differential equation

$$\frac{d^3 c}{dt^3} + 3 \frac{d^2 c}{dt^2} + 7 \frac{dc}{dt} + 5c = \frac{d^2 r}{dt^2} + 4 \frac{dr}{dt} + 3r.$$

Q3. Find the transfer function relating the capacitor voltage, $V_c(s)$, to the input voltage, $V(s)$ as shown in Figure 1.

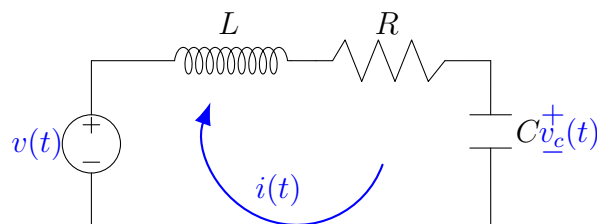


Figure 1: Series RLC circuit.

Q4. Find the transfer function for the circuit given in Figure 2.

Q5. Find the transfer function, $X(s)/F(s)$, for the system of Figure 3.

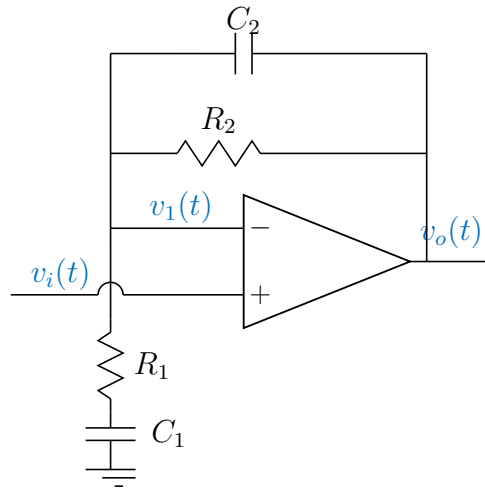


Figure 2: Op-amp RC circuit with standard parallel-plate capacitors.

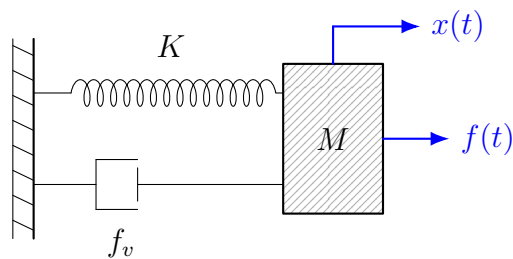


Figure 3: Mass-spring-damper system.